

CIS530 M3: LoRA-Based Extension for Multilingual Toxic Comment Detection

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Project context. We investigate multilingual toxic comment detection using the Jigsaw 2020 dataset, aiming to identify toxic user comments in Spanish, Italian, and Turkish. In Milestone 2, our strong Detoxify (XLM-R) baseline achieved high ROC-AUC but suffered from severely low recall in Spanish and uneven error patterns across languages. To address this, Milestone 3 introduces a parameter-efficient LoRA adapter combined with class-weighted BCE to improve recall while maintaining lightweight training.

Extension Description

We extend the Detoxify/XLM-R[1] baseline using **parameter-efficient LoRA**[2], freezing the backbone and injecting low-rank update matrices into the query and value projections of each attention block. With rank $r = 8$, scaling $\alpha = 16$, and dropout 0.1, this setup introduces only $\sim 1\text{M}$ trainable parameters per language, reducing training cost by over 99% while enabling lightweight, language-specific adaptation.

To address the severe label imbalance in Jigsaw 2020, we apply a **class-weighted BCEWithLogitsLoss** using $\text{pos_weight} = \frac{N_{\text{neg}}}{N_{\text{pos}}}$ for each language. This increases the influence of toxic examples and directly targets the low-recall failure mode observed in Milestone 2, improving sensitivity to minority-class toxicity while keeping computation efficient.

Empirical Evaluation

We evaluate on the held-out Spanish, Italian, and Turkish test sets. Metrics follow the main project (ROC-AUC + F1 for toxic class):

Language	Model	F1	ROC-AUC	Precision	Recall	FPR
es	Detoxify (M2)	0.615	0.918	0.870	0.476	0.099
	LoRA + BCE	0.398	0.832	0.250	0.976	0.589
it	Detoxify (M2)	0.558	0.876	0.410	0.878	0.094
	LoRA + BCE	0.371	0.666	0.275	0.571	0.366
tr	Detoxify (M2)	0.822	0.982	0.732	0.938	0.019
	LoRA + BCE	0.606	0.914	0.588	0.625	0.052

Table 1: Comparison of our LoRA extension with the Detoxify multilingual XLM-R strong baseline.

Table 1 reports test-set performance on es, it, and tr. The LoRA + BCE extension substantially improves recall on the minority toxic class—most notably for Spanish ($0.476 \rightarrow 0.976$)—demonstrating that class-weighted training successfully corrects the recall failure identified in Milestone 2. This improvement comes at the expected cost of precision and overall F1, reflecting a shift toward more sensitive toxicity detection under severe class imbalance.

Discussion

While our LoRA + class-weighted BCE extension successfully achieves its primary objective of improving toxic-class recall—especially for Spanish, where recall rises from 0.476 to 0.976—this gain comes with notable trade-offs in precision and overall F1. These results highlight the difficulty of balancing sensitivity and specificity under extreme class imbalance, and suggest that future work may require threshold tuning, focal-loss variants, or multi-objective training to recover precision without losing recall. Nonetheless, the experiment demonstrates that parameter-efficient methods can meaningfully shift model behavior while keeping computation lightweight.

References

- [1] A. Conneau and K. . Khandelwal. Unsupervised cross-lingual representation learning at scale. In *ACL*, 2020.
- [2] E. Hu, Y. Shen, P. Wallis, and Z. . Allen-Zhu. Lora: Low-rank adaptation of large language models. In *ICLR*, 2022.